Speech Recognition For Green Technology

Group 5

Green Technology For Sustainable Development

Group 5

Green Technology For Sustainable Development

01 Introduction to green technology
02 Need for sustainable development
03 Green buildings and smart cities
04 Artificial photosynthesis in green technology
05 Green data centers using AI
06 Green technology in agriculture and food security
07 Social, economic perspectives
08 Climate adaptation technologies
09 Green Transport and mobility innovations
10 Challenges and future directions
11 Conclusion

Introduction to Green Technology

01
02
03
04
05
06
07
08
09
10
11

- ❖Green Technology – eco-friendly innovations
- ❖Reduces environmental impact
- ❖Promotes renewable energy use
- ❖Sustainable Development – future-focused growth
- ❖Meets present needs responsibly
- ❖Balances economy and environment
- ❖Green technology supports sustainability

01
02
03
04
05
06
07
08
09
10
11

- Conservation of Natural Resources
- Environmental Protection
- Economic Stability
- Social Equality
- Reduction of Climate Change Effects
- Protection of Biodiversity
- Long-Term Development

Need for sustainable development

WITH GREAT POWER COMES
GREAT RESPONSIBILITY
- SOURAV

01
02
03
04
05
06
07
08
09
10
11

Smart Cities

- Economic Sustainability
- Environmental Sustainability
- Social Sustainability

Green Buildings

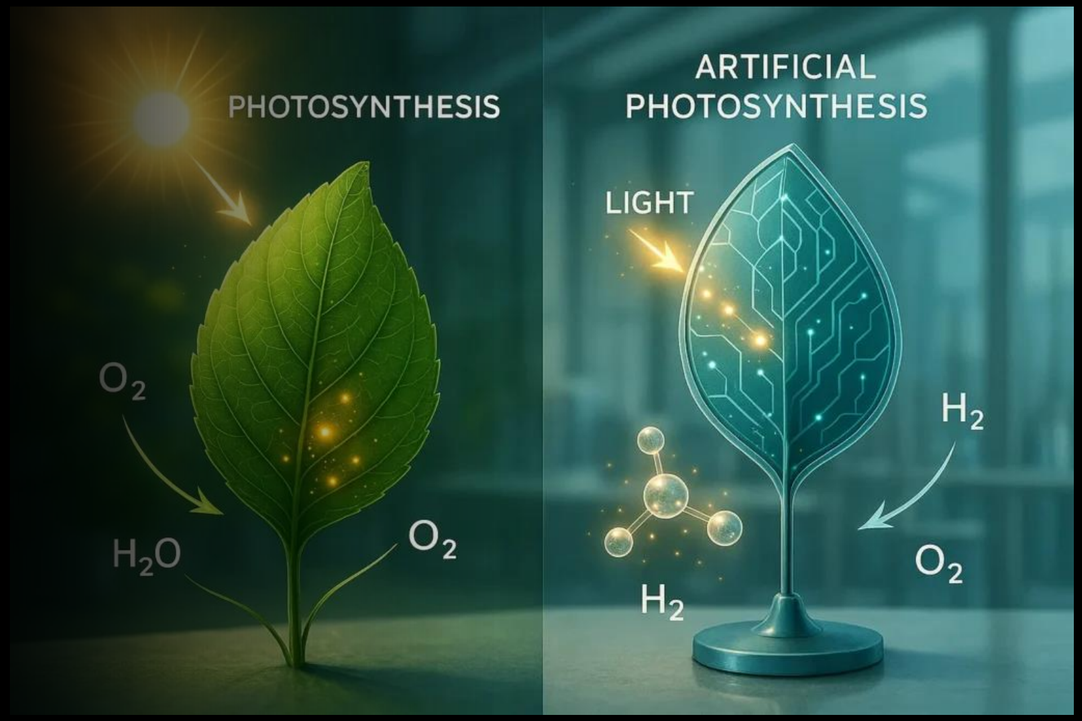
- Economic Pillar
- Environmental Pillar
- Social Pillar

Green Buildings and Smart Cities

Artificial photosynthesis in Green Technology

01
02
03
04
05
06
07
08
09
10
11

- ❖Copying Nature
- ❖Storing Sunlight
- ❖Cleaning the Air
- ❖Better than Plants
- ❖Green Fuel for Big Machines



Artificial photosynthesis in Green Technology

01
02
03
04
05
06
07
08
09
10
11

Applications	Major Uses
❖Green Hydrogen Production	❖Decentralized Fuel Refineries
❖Synthetic Hydrocarbons	❖Industrial Carbon Recycling
❖Carbon-Neutral Plastics	
❖ Green Ammonia	
❖Sustainable Fragrances & Fine Chemicals:	

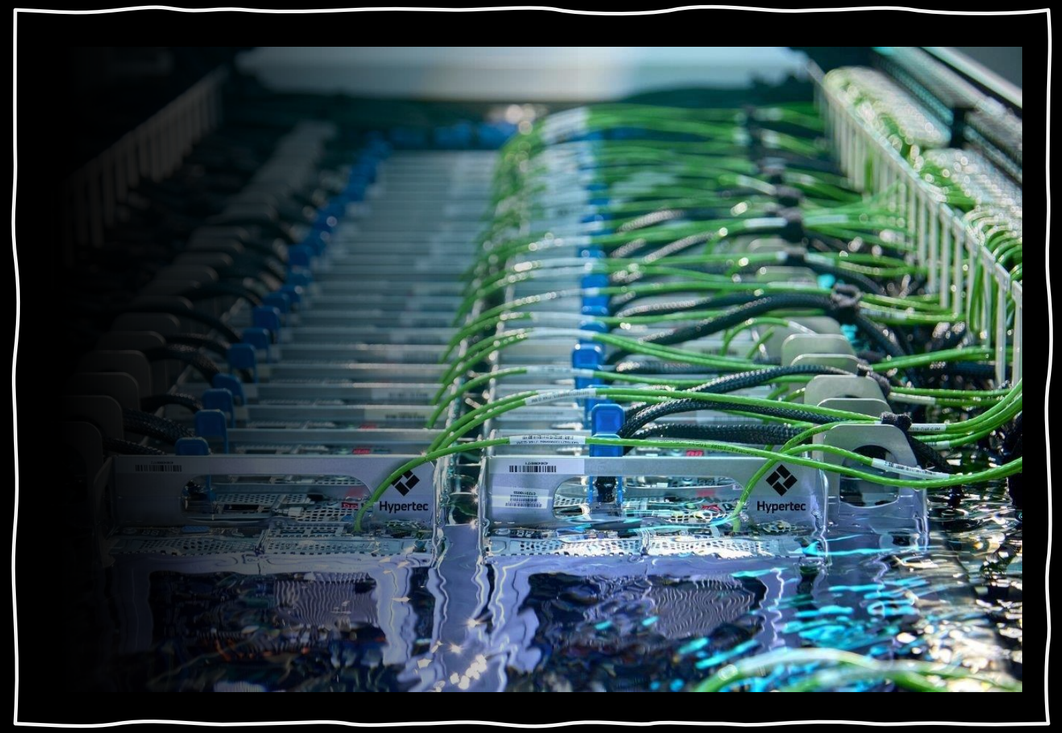
Artificial photosynthesis in Green Technology

01
02
03
04
05
06
07
08
09
10
11

Advantages	Disadvantages
❖Energy Storage	❖High Initial Cost
❖Carbon Negative	❖Low Market Maturity
❖High Energy Density	
❖Superior Efficiency	
❖Land Use	

Green data centers using AI

01
02
03
04
05
06
07
08
09
10
11



❖ Introduction

❖ Energy-Efficient Cooling

❖ Smart Power Management

❖ Renewable Energy Integration

❖ Predictive Maintenance

❖ Environmental Benefits

❖ Real-World Examples

01
02
03
04
05
06
07
08
09
10
11

Green technology in agriculture and Food Security

- ❖ Precision Farming
- ❖ Indoor Growing
- ❖ Stronger Crops
- ❖ Saving Resources
- ❖ Reducing Waste

Social & Economic Perspectives

01
02
03
04
05
06
07
08
09
10
11

❖ Introduction
❖ A Historical Perspective
❖ Job Creation
❖ Impact on Poor Communities
❖ Health & Social Benefits
❖ Government & Policy Support
❖ Conclusion

Climate Adaptation Technologies

Climate change causes:

- ❖Heavy rainfall and floods
- ❖Extreme heat
- ❖Drought and water shortage
- ❖Strong storms

Adaptation technologies :

- ❖Flood barriers to protect cities
- ❖Cool roofs to reduce heat
- ❖Drip irrigation to save water
- ❖Rainwater harvesting systems

Why It Is Important?

- ❖Reduces damage from floods and storms
- ❖Helps farmers grow crops during drought
- ❖Protects people from extreme heat
- ❖Saves and manages water properly
- ❖Protects houses, roads, and buildings
- ❖Reduces risk to human life

Climate Adaptation Technologies

Green Transport and mobility innovations

❖Concept

❖Environment-friendly transportation system
❖Reduces air pollution and carbon emissions
❖Uses clean energy sources
❖Promotes sustainable mobility

❖Key Innovations:

❖ Electric Vehicles (EVs)
❖ Hydrogen fuel vehicles
❖ Biofuel-based transport
❖ Smart public transport systems like Metro

Green Transport and mobility innovations

❖Impact & Benefits:

❖Improves air quality
❖Saves energy
❖Reduces fuel dependency
❖Lowers traffic congestion
❖Supports sustainable development

❖Future of Green Mobility:

❖ Autonomous electric vehicles
❖ Electric air taxis
❖ Advanced battery technology
❖ Smart transport networks

Challenges & Future directions

❖Challenges

❖High initial cost
❖Lack of awareness
❖Technological Limitations
❖Infrastructure Problems

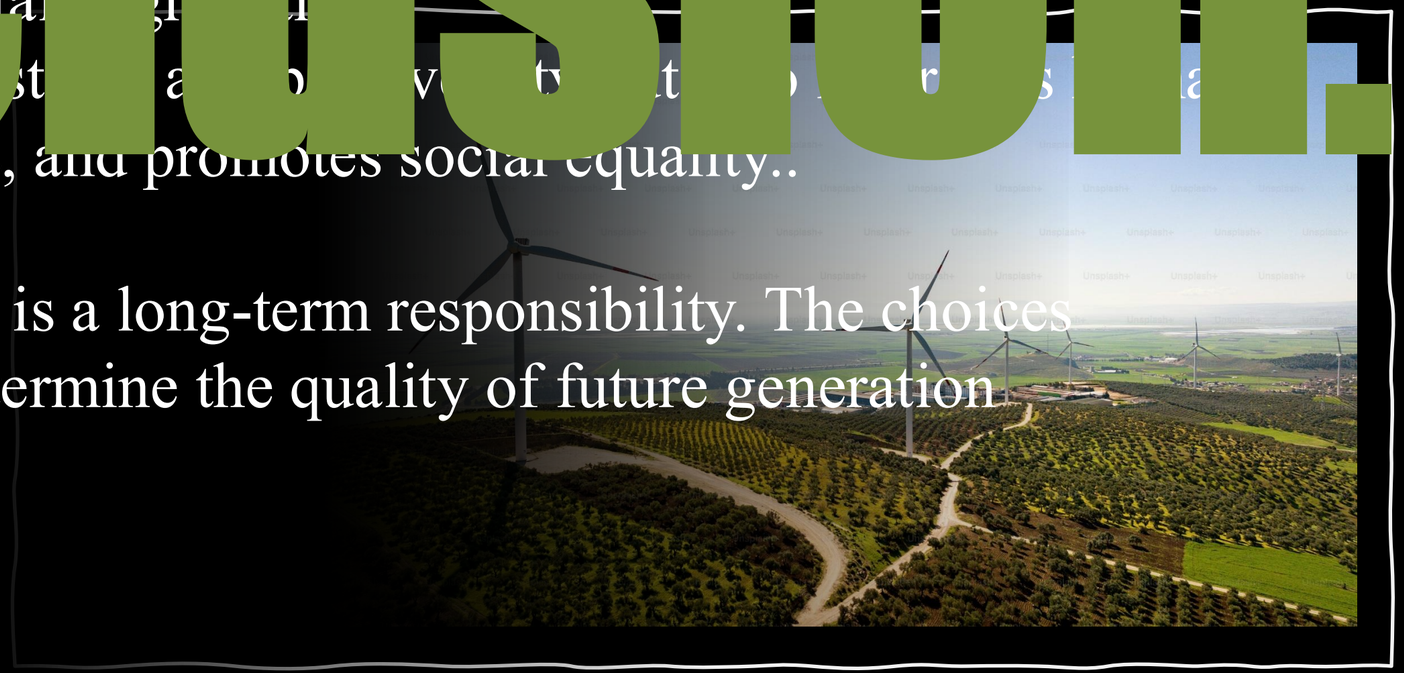
❖Future Directions

❖ Improvement in renewable Energy
❖ Green Hydrogen Energy
❖ Electric Vehicle Growth
❖ Waste to Energy Technology

01
02
03
04
05
06
07
08
09
10
11

Conclusion.

- In conclusion, green technology plays a crucial role in achieving sustainable development by creating a balance between environmental protection, economic growth, and social well-being.
- [illegible] technologies [illegible] reduce pollution, decarbonize emissions, [illegible] energy efficiency, [illegible] development [illegible] economic [illegible] environmental [illegible] [illegible] health, creates green jobs, and promotes social equality..
- Sustainable development is a long-term responsibility. The choices we make today—will determine the quality of future generation.



Conclusion.

- In conclusion, green technology plays a crucial role in achieving sustainable development by creating a balance between environmental protection, economic growth, and social well-being.
- These technologies help reduce pollution, lower carbon emissions, and improve energy efficiency, ensuring that development does not come at the cost of environmental degradation.
- It not only protects ecosystems and biodiversity but also improves human health, creates green jobs, and promotes social equality..

- Sustainable development is a long-term responsibility. The choices we make today—will determine the quality of future generation

WITH GREAT POWER COMES GREAT RESPONSIBILITY

- SOURAV